

Expt 9	Adaptive Filtering for Echo Cancellation
Date:	9.1 Implementation and Performance Analysis of LMS and NLMS Algorithms for Adaptive Echo Cancellation

Code:

```

clc;
clear;
close all;

%% Input Signal
N = 1000;
x = randn(1,N);           % Original input signal

%% Echo Signal
h = [0.8 0.5 0.3];       % Echo path
d = filter(h,1,x);        % Desired (echo) signal

%% Adaptive Filter Parameters
M = 3;                   % Filter length
mu1 = 0.05;              % LMS step size
mu2 = 0.8;               % NLMS step size

%% Initialization
wLMS = zeros(M,1);
wNLMS = zeros(M,1);

eLMS = zeros(1,N);
eNLMS = zeros(1,N);

yLMS = zeros(1,N);
yNLMS = zeros(1,N);

delta = 1e-6;            % Avoid division by zero

%% LMS Algorithm
for n = M:N

    u = x(n:-1:n-M+1)';

    % Filter output
    yLMS(n) = wLMS' * u;

    % Error
    eLMS(n) = d(n) - yLMS(n);

    % Weight update

```

```
wLMS = wLMS + mu1 * eLMS(n) * u;
```

```
end
```

```
%% NLMS Algorithm
```

```
for n = M:N
```

```
    u = x(n:-1:n-M+1)';
```

```
    % Filter output
```

```
    yNLMS(n) = wNLMS' * u;
```

```
    % Error
```

```
    eNLMS(n) = d(n) - yNLMS(n);
```

```
    % Weight update
```

```
    normFactor = delta + (u' * u);
```

```
    wNLMS = wNLMS + (mu2 / normFactor) * eNLMS(n) * u;
```

```
end
```

```
%% Figure 1 : Input and Echo Signal
```

```
figure;
```

```
plot(x,'b');
```

```
hold on;
```

```
plot(d,'r');
```

```
grid on;
```

```
legend('Input Signal','Echo Signal');
```

```
title('Input and Echo Signal');
```

```
xlabel('Samples');
```

```
ylabel('Amplitude');
```

```
%% Figure 2 : LMS Output
```

```
figure;
```

```
plot(yLMS,'b');
```

```
grid on;
```

```
title('LMS Filter Output');
```

```
xlabel('Samples');
```

```
ylabel('Amplitude');
```

```
%% Figure 3 : NLMS Output
```

```
figure;
```

```
plot(yNLMS,'r');
```

```
grid on;
```

```
title('NLMS Filter Output');
```

```
xlabel('Samples');
```

```
ylabel('Amplitude');
```

```
%% Figure 4 : MSE Comparison
```

```
figure;
```

```

semilogy(eLMS.^2,'r','LineWidth',1.5);
hold on;
semilogy(eNLMS.^2,'b','LineWidth',1.5);
grid on;
legend('LMS','NLMS');
title('Mean Square Error Comparison');
xlabel('Iterations');
ylabel('Squared Error');

```

Output:

Fig 1:

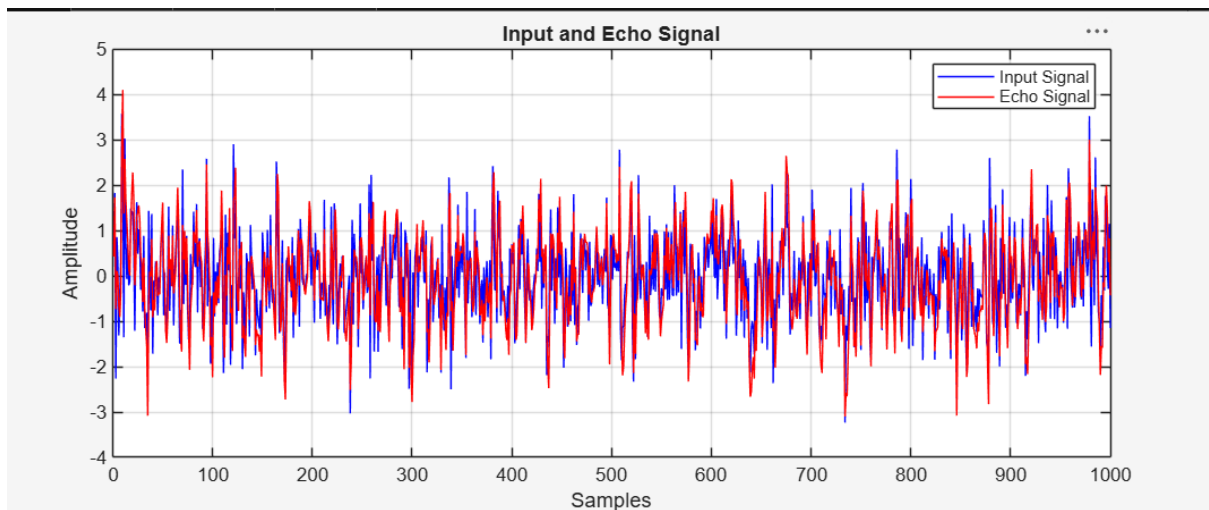


Fig 2:

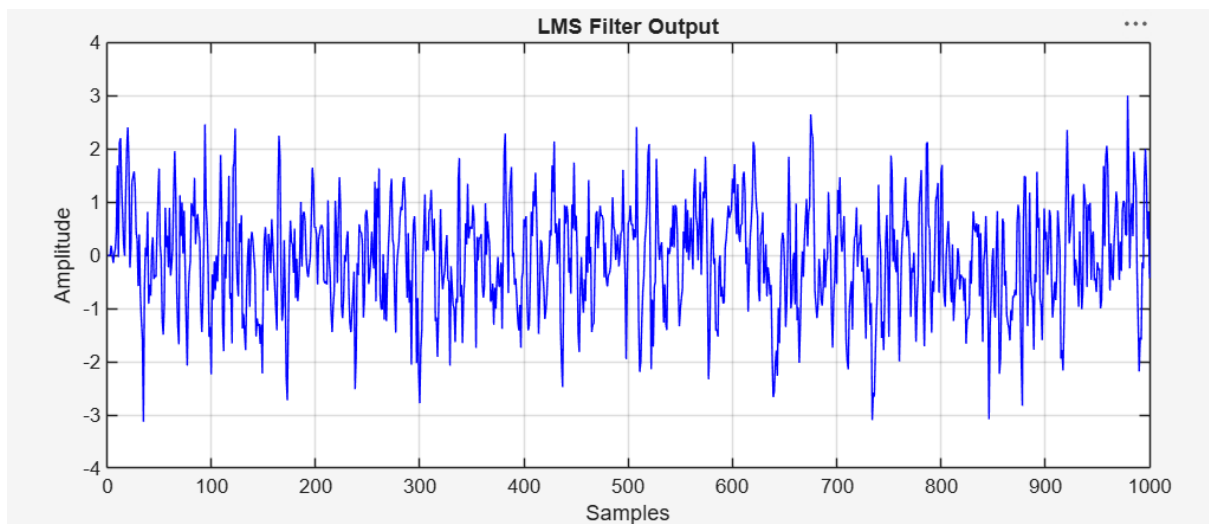


Fig 3:

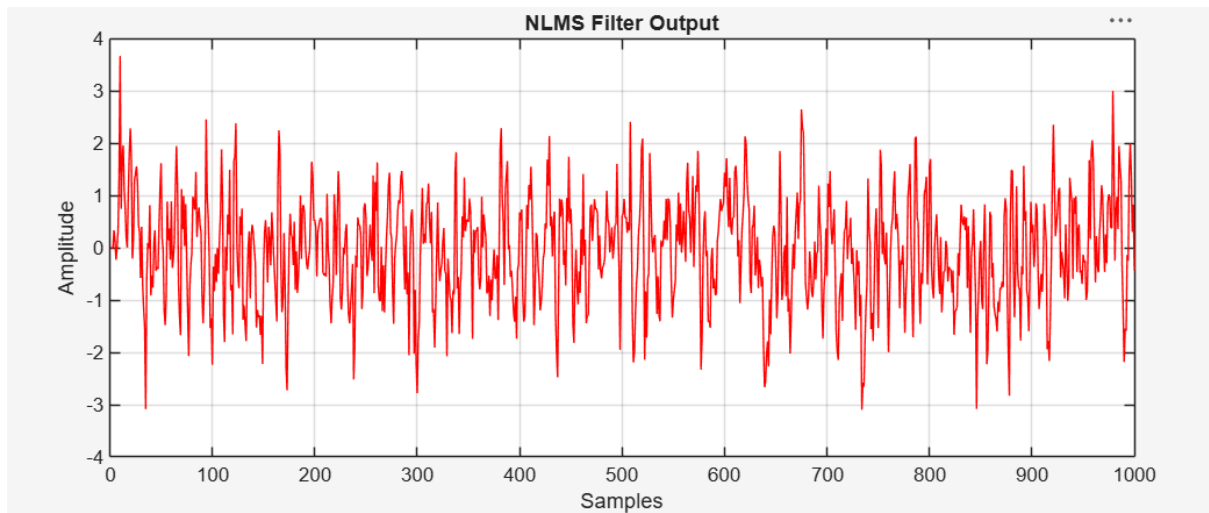


Fig 4:

